

## Khoi Dang Tran

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### EDUCATION

**Case Western Reserve University** | 2025 - Present  
PhD in Pathology

**Case Western Reserve University** | 2021 - 2025  
Bachelor of Arts in Biochemistry |  
GPA: 3.916/4.0

### RESEARCH EXPERIENCE

**Case Western Reserve University** *August 2022 - Present*

Undergraduate Research Assistant under Dr. Alexa Woo

$\beta$ -arrestin2 knock-out in cells and *in vivo*

- Examined the effects of  $\beta$ -arrestin2 knock-out on  $\alpha$ -synuclein levels in transgenic mice by immunoblot

$\beta$ -arrestin2 and LMBRD2 interaction in  $\alpha$ -synuclein pathogenesis

- Examined LMBRD2 expression levels and effects on cell viability using immunoblot, immunocytochemistry, and immunohistochemistry
- Performed stereotaxic injection of  $\alpha$ -synuclein PFFs into mice brain to study LMBRD2-mediated  $\alpha$ -synuclein pathogenesis
- Examined motor and cognitive function of  $\alpha$ -synuclein PFFs injected mice by performing behavioral testing

CHCHD2 dysfunction in Parkinson's Disease

- Optimized cathepsin activity assay to study changes in protease activity in transgenic mice brain tissues
- Used *in situ* Proximity Ligation Assay (PLA) to study localization of CHCHD2-lysosomal cathepsin interaction
- Characterized lysosomal cathepsin expression and activity changes in CHCHD2-T61I mutation using immunoblot, immunohistochemistry, and fluorescence microscopy

SSH1 inhibitors to mitigate Alzheimer's Disease (AD)

- Used an *in vitro* assay to screen for potent SSH1 inhibitors amongst 2700 drugs from a FDA-approved drug library
- This project is part of my Senior Thesis

### SKILLS and TECHNIQUES

- DNA cloning and purification
- Protein isolation and purification

- Western blotting, immunocytochemistry, and immunohistochemistry
- Proximity ligation assay and analysis on cells and tissues
- Fluorescence microscopy
- Mammalian cell culture
- Mouse brain and organs harvesting
- Stereotaxic surgery and injection
- Motor and cognitive behavioral testing on mice

## LEADERSHIP EXPERIENCE

### Vietnamese Student Association – Secretary

*August 2023 – May 2024*

Case Western Reserve University

- Manages all correspondence and keep important minutes of all meetings
- Drafts and sends weekly newsletter to active members.
- Organizes weekly executive meetings and meetings with collaborating organizations.

## PUBLICATIONS

T.R. Kee, B.M. Masters, **K.D. Tran**, H. Jeon, S.A. Khan, E. Vaquez-Rosa, P.S. Sridharan, H. Fujioka, D. Chaput, X. Wang, X. Zhao, A.A. Pieper, D.E. Kang, & J.A. Woo. Disruption of the mitochondria-lysosome axis by CHCHD2 dysfunction drives proteinopathy associated with brain aging and Lewy body disease. In preparation for Submission in 12/2024.

## ABSTRACTS

**K.D. Tran**, T.R. Kee, B.M. Masters, and J.A. Woo. Parkinson's Disease–Associated CHCHD2–T61I Promotes Proteinopathy Via Lysosomal Cathepsin Inhibition *in vivo*. Abstracts of the CWRU Intersection Poster Presentation, 2024.

## HONORS AND AWARDS

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|-------------|-----------------------------------------------------------------------------------------------------|
| 2024        | Provost Summer Undergraduate Research Scholarship (PSURG)<br><i>Case Western Reserve University</i> |
| 2021 - 2024 | Dean's High Honor List, <i>Case Western Reserve University</i>                                      |
| 2021        | Dean's Honor List, <i>Case Western Reserve University</i>                                           |